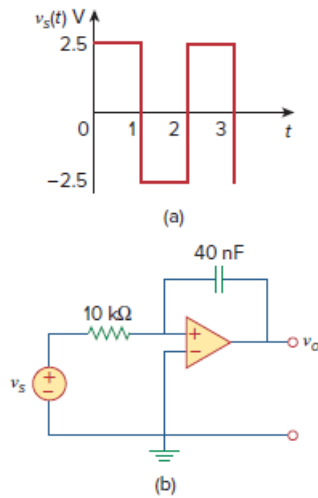


Problem

The square wave in Fig. 17.78(a) is applied to the circuit in Fig. 17.78(b). Find the Fourier series of $v_o(t)$.

Figure 17.78:



Step-by-step solution

Step 1 of 4

Refer to table 17.3 in the textbook.

Write the expression for Fourier series of the square wave.

$$f(t) = \frac{4A}{\pi} \sum_{n=1}^{\infty} \left(\frac{1}{2n-1} \sin(2n-1)\omega_0 t \right) \dots\dots (1)$$

Refer to Figure 17.78 (a) in the textbook for square wave.

The amplitude of the square wave is, $A = 2.5$

The time period of square wave is, $T = 2$

Determine the angular frequency.

$$\begin{aligned} \omega_0 &= \frac{2\pi}{T} \\ &= \frac{2\pi}{2} \\ &= \pi \end{aligned}$$

Step 2 of 4

Substitute 2.5 for A and π for ω_0 .

$$f(t) = \frac{10}{\pi} \sum_{n=1}^{\infty} \left(\frac{1}{2n-1} \sin(2n-1)\omega_0 t \right)$$

Refer to Figure 17.78(b) in the textbook.

Apply Kirchhoff's current law at noninverting terminal.

$$\frac{0 - v_s}{10 \text{ k}\Omega} + \frac{0 - v_o}{\left(\frac{1}{j\omega(40 \times 10^{-9})} \right)} = 0$$

$$v_o j\omega(40 \times 10^{-9}) = -\frac{v_s}{10 \text{ k}\Omega}$$

$$v_o = -\frac{v_s}{j\omega 10^4 (40 \times 10^{-9})}$$

$$v_o = -\frac{v_s}{j\omega(4 \times 10^{-4})}$$

$$v_o = \frac{jv_s}{\omega(4 \times 10^{-4})}$$

Step 3 of 4

The input signal is,

$$v_s(t) = \frac{10}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \sin(2n-1)\pi t$$

The frequency of input signal is, $\omega = (2n-1)\pi$

Substitute $\omega = (2n-1)\pi$ in the output expression.

$$\begin{aligned} v_o &= \frac{jv_s}{(2n-1)\pi(4 \times 10^{-4})} \\ &= \left(\frac{10^4}{4\pi(2n-1)} \angle 90^\circ \right) v_s \end{aligned}$$

Step 4 of 4

Substitute $\frac{10}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \sin(2n-1)\pi t$ for $v_s(t)$.

$$v_o = \left(\frac{10^4}{4\pi(2n-1)} \angle 90^\circ \right) \left(\frac{10}{\pi} \sum_{n=1}^{\infty} \frac{1}{2n-1} \sin(2n-1)\pi t \right)$$

Magnitude multiplied and phase is added.

$$\begin{aligned} v_o &= \frac{10^5}{4\pi^2} \left(\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \sin[(2n-1)\pi t + 90^\circ] \right) \\ &= \frac{10^5}{4\pi^2} \left(\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos(2n-1)\pi t \right) \end{aligned}$$

Therefore, the expression of $v_o(t)$ is $\boxed{\frac{10^5}{4\pi^2} \left(\sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \cos(2n-1)\pi t \right)}$.